Statistical Word Segmentation: Emergent Structure in a Next-Character RNN

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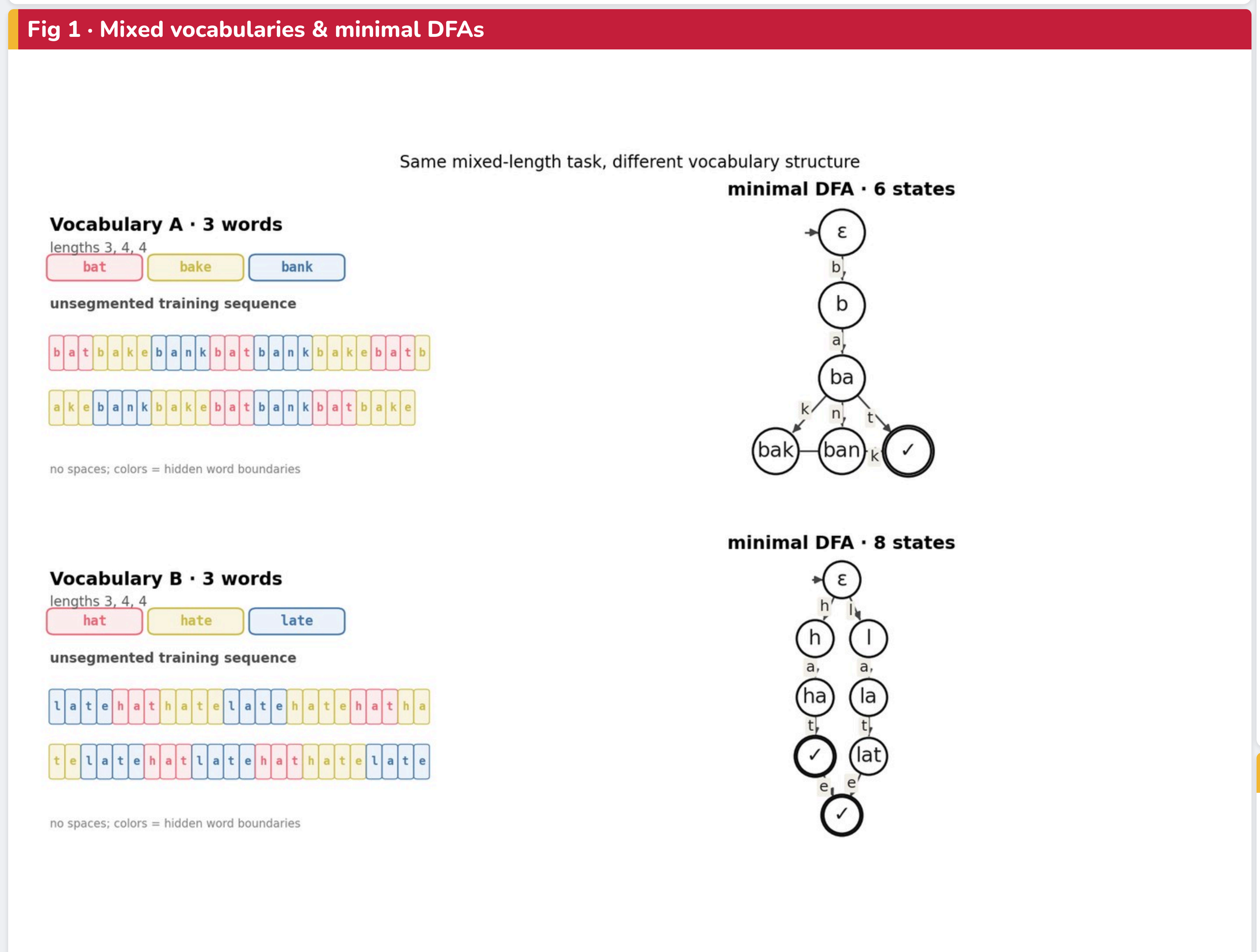
Code & figures on GitHub

Abstract

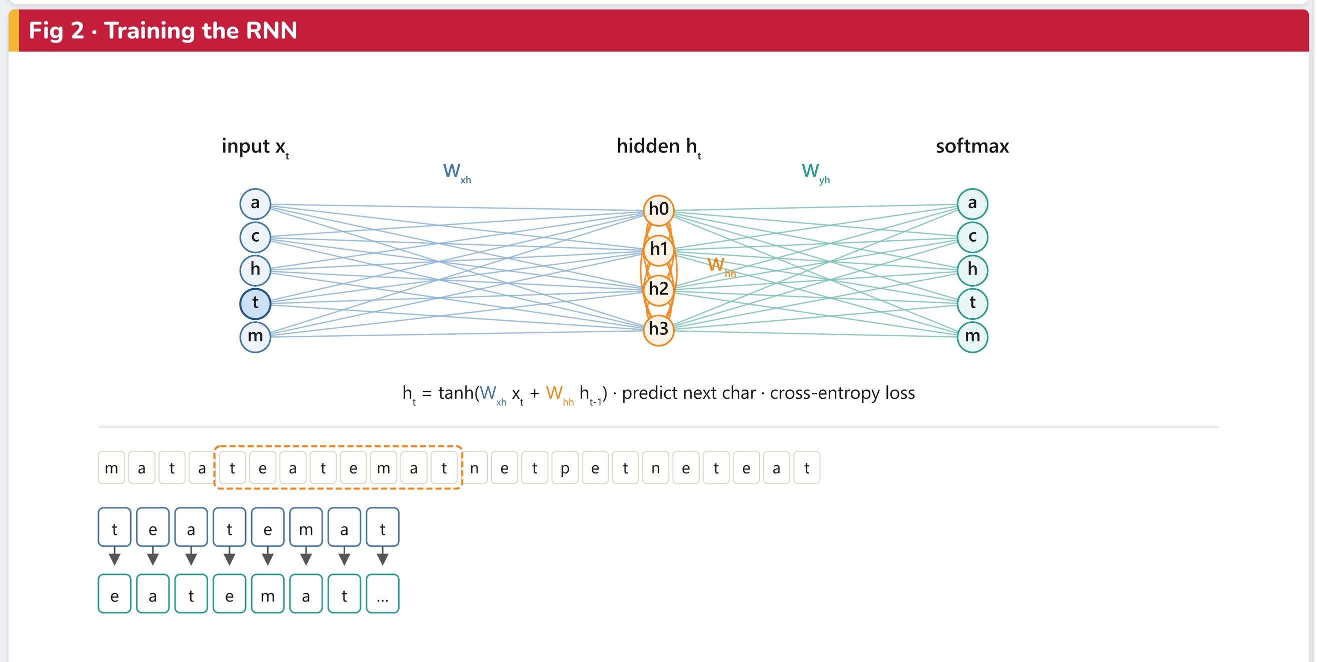
Eight-month-old infants can segment continuous speech by tracking transitional probabilities (Saffran, Aslin & Newport, 1996). We ask whether a vanilla Elman RNN trained only on next-character prediction develops internal representations aligned with word structure. After learning, the network generates legal vocabulary items, and its hidden states become a continuous embedding of the vocabulary's minimal DFA. On an eight-word *-ate/-at* demo, DFA state explains $\eta^2 \approx 0.88$ of condensed hidden variance; word identity is much weaker ($\eta^2 \approx 0.06$). Fifty mixed-length English vocab runs ($H{=}100$) show that dimensionality, training cost, and readouts track minimized DFA size. Weight matrices become letter-columnar in W_{xh} and locally clumped in W_{hh}, most clearly for small automata.

Fig 1 · Mixed vocabularies & minimal DFAs



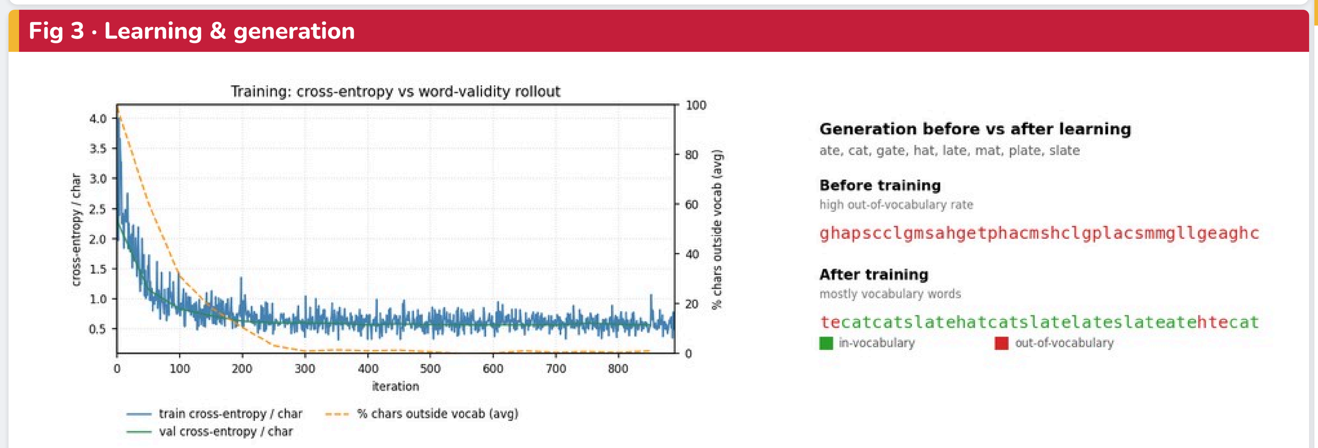
Two ≤3-word conditions (3–4 letter words) with long unsegmented streams and prefix-labeled minimal DFAs (6 vs 8 states).

Fig 2 · Training the RNN



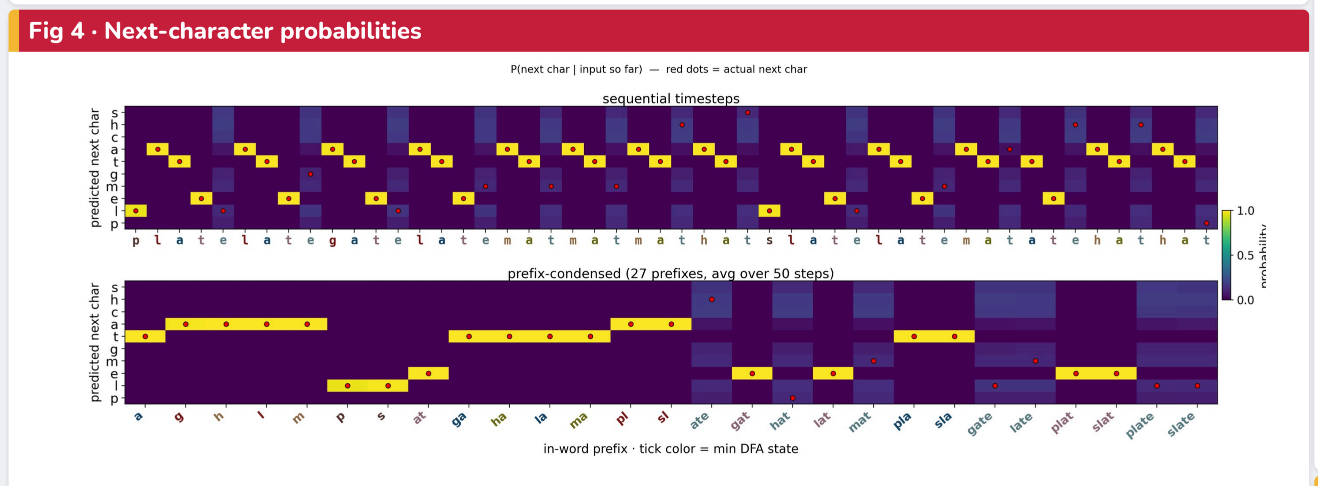
Character-level Elman RNN. Input x_t predicts the next character via $h_t = \tanh(W_{xh}x_t + W_{hh}h_{t-1})$; cross-entropy loss.

Fig 3 · Learning & generation



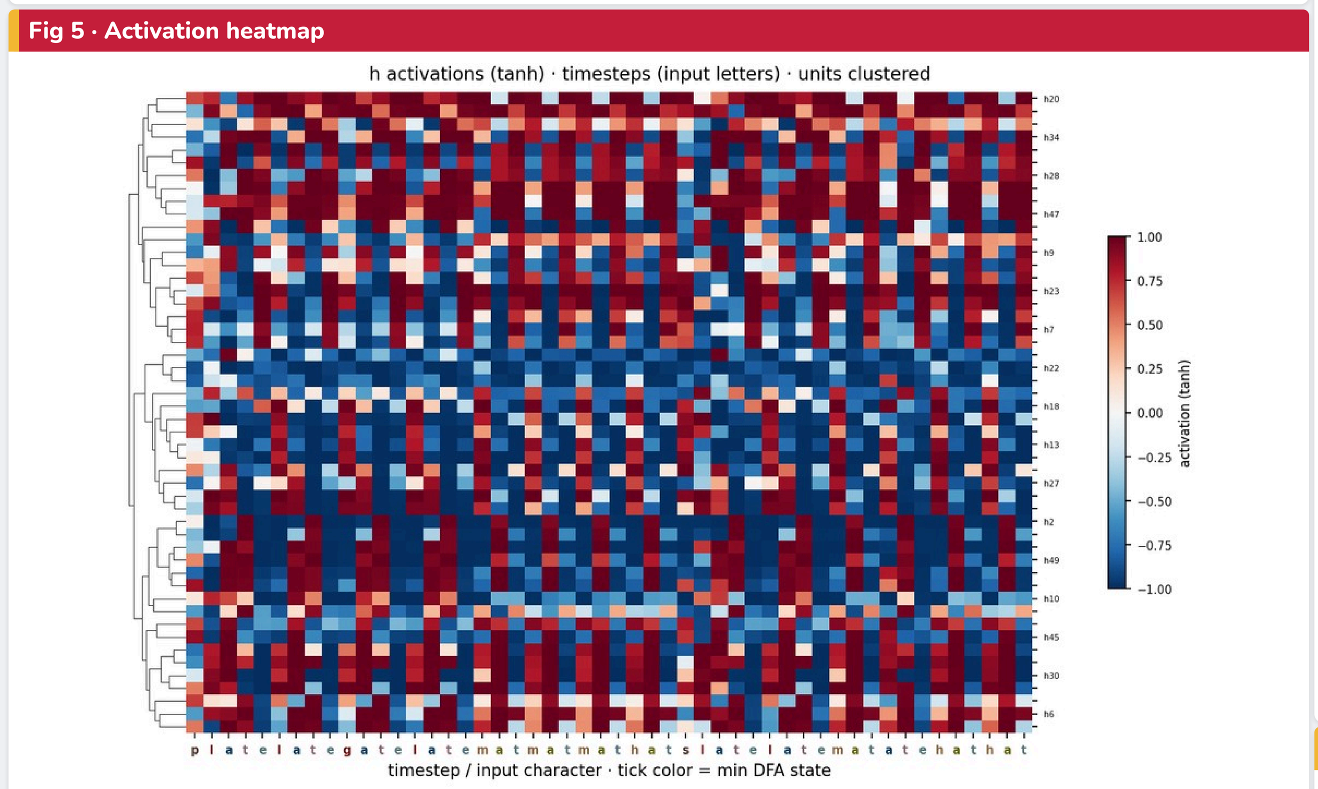
Loss and OOV collapse together; rollouts become legal vocabulary words.

Fig 4 · Next-character probabilities



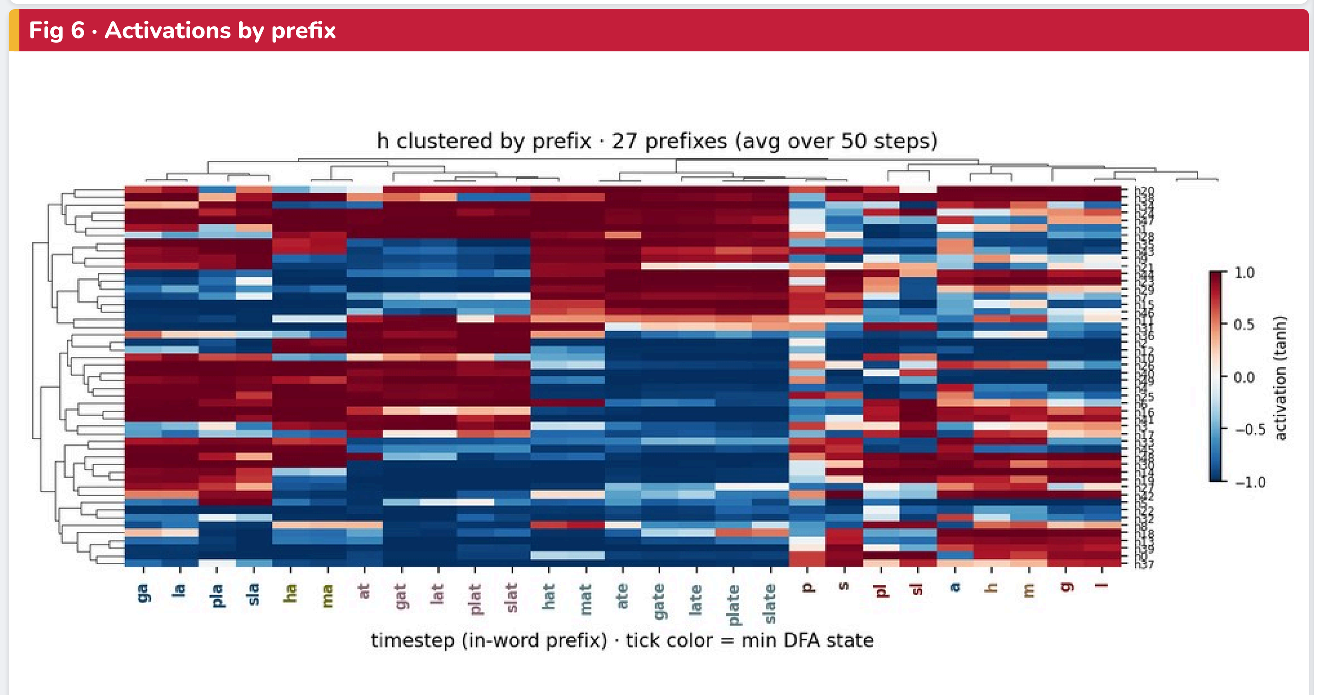
Softmax $P(\text{next char} \mid \text{input so far})$. Red dots = actual next character.

Fig 5 · Activation heatmap



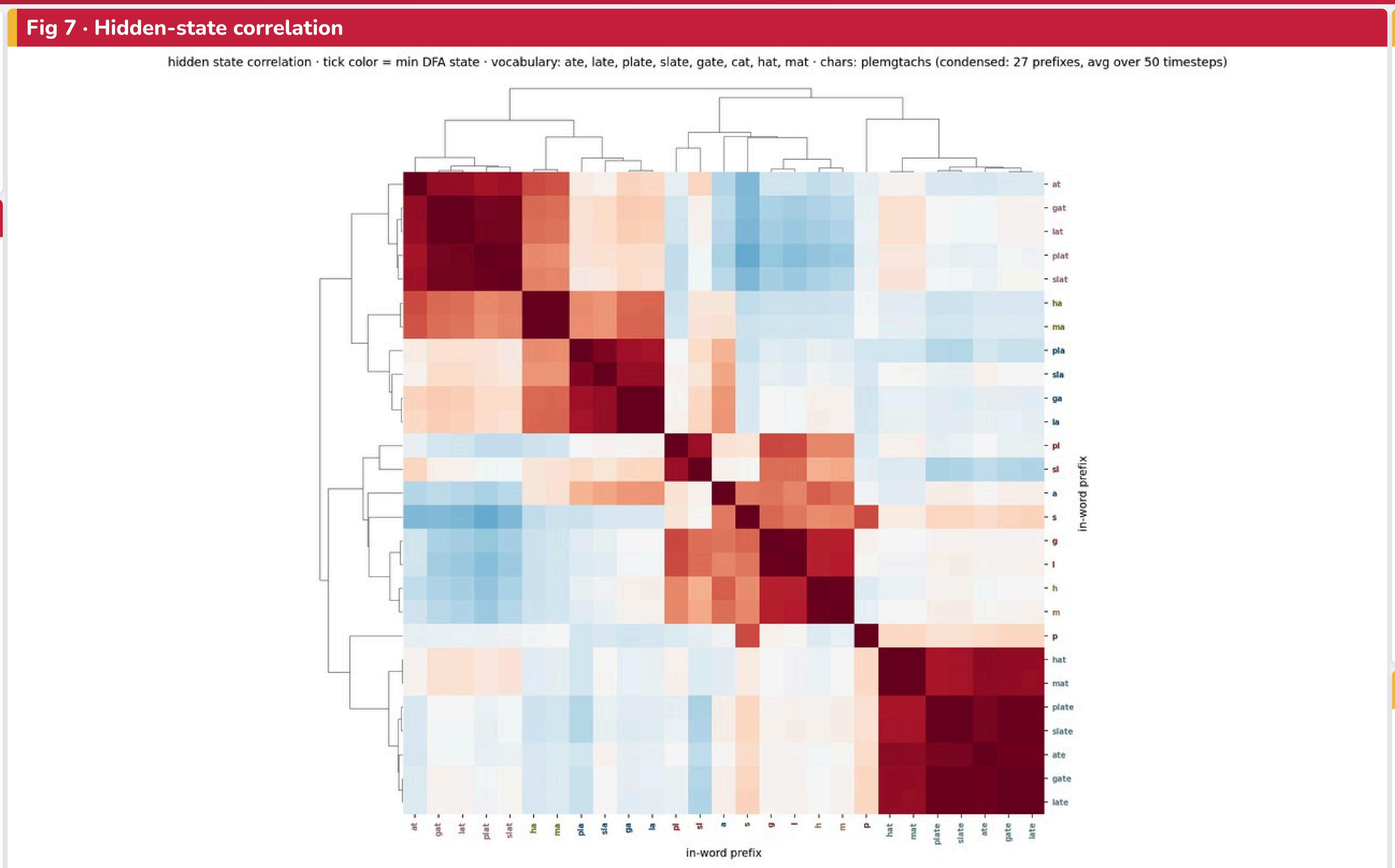
Hidden units over timesteps (input letters; units clustered).

Fig 6 · Activations by prefix



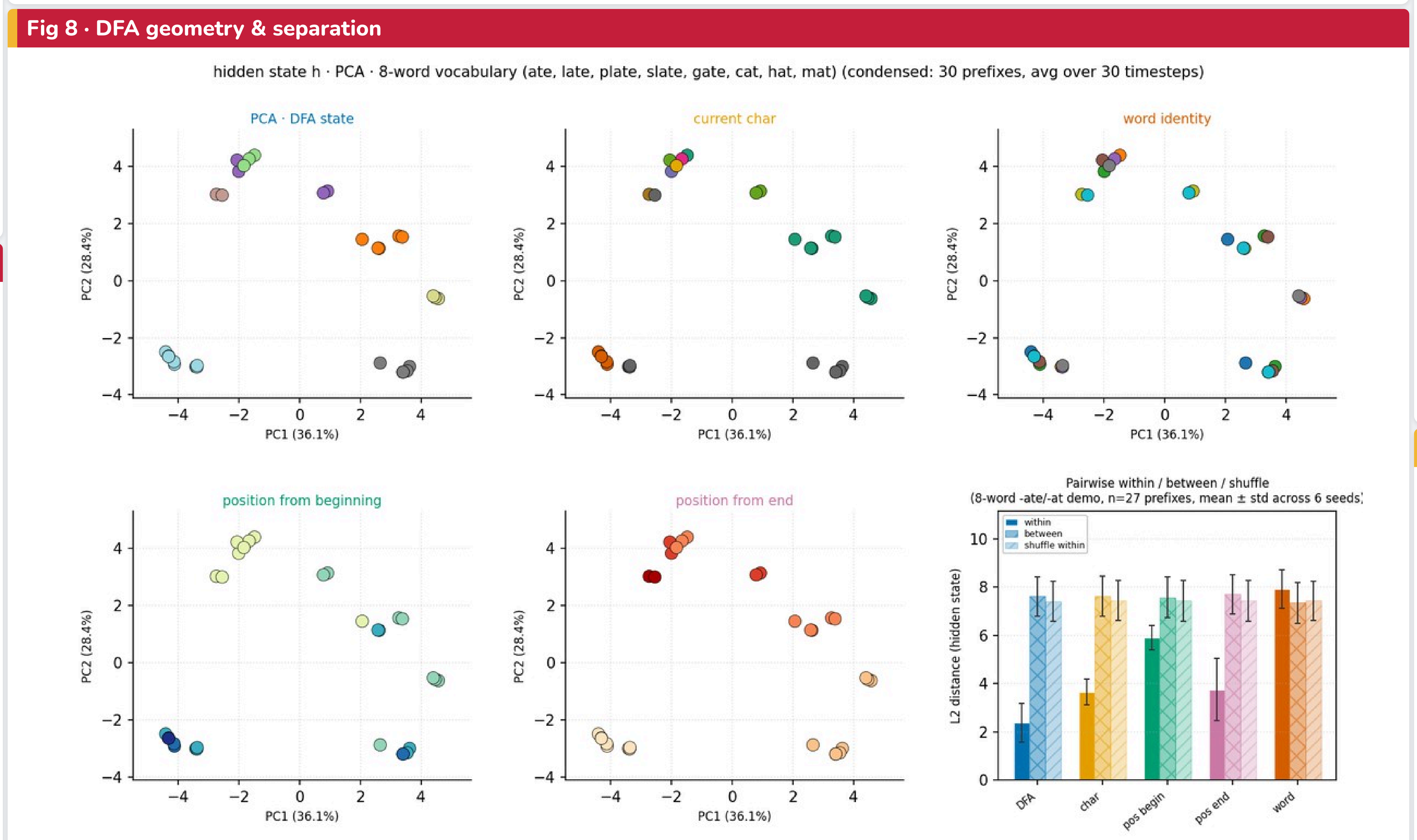
Activations clustered by in-word prefix. Tick color = DFA state.

Fig 7 · Hidden-state correlation



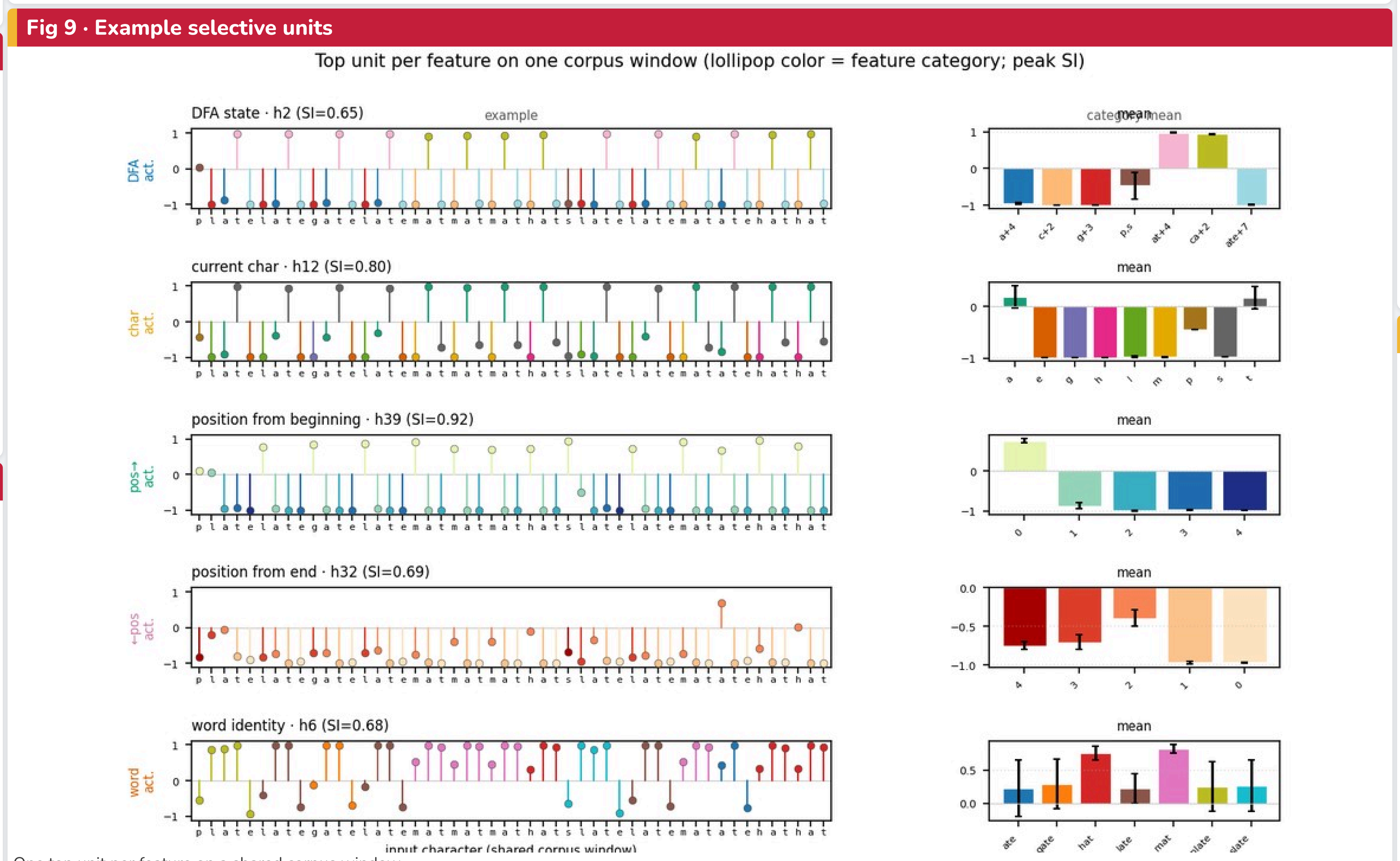
Prefix × prefix correlation. Blocks merge prefixes that share a future.

Fig 8 · DFA geometry & separation



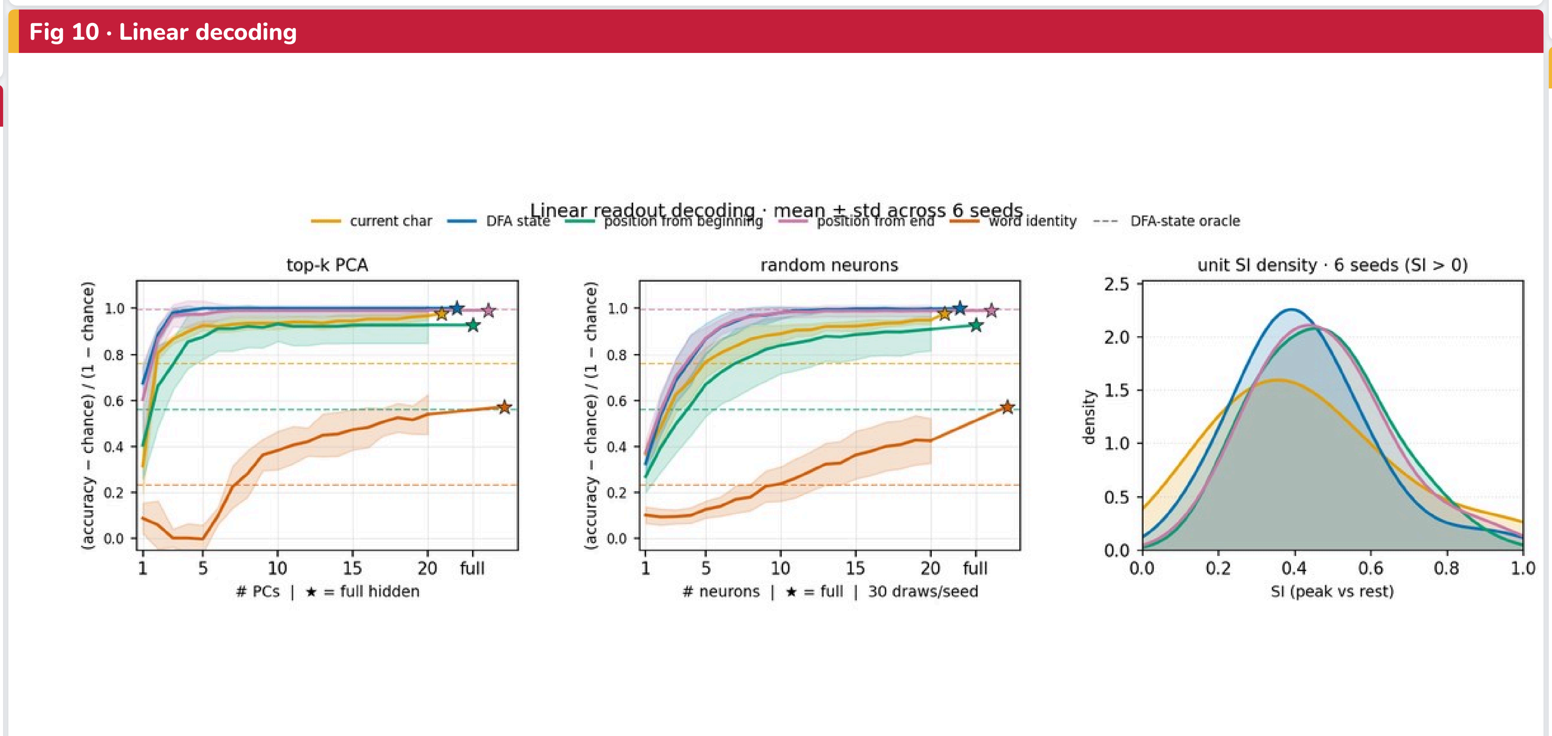
PCA of $\mathbf{h}$ by feature; pairwise within/between/shuffle at end of row. DFA $\eta^2 \approx 0.88$; word identity ≈ 0.06.

Fig 9 · Example selective units



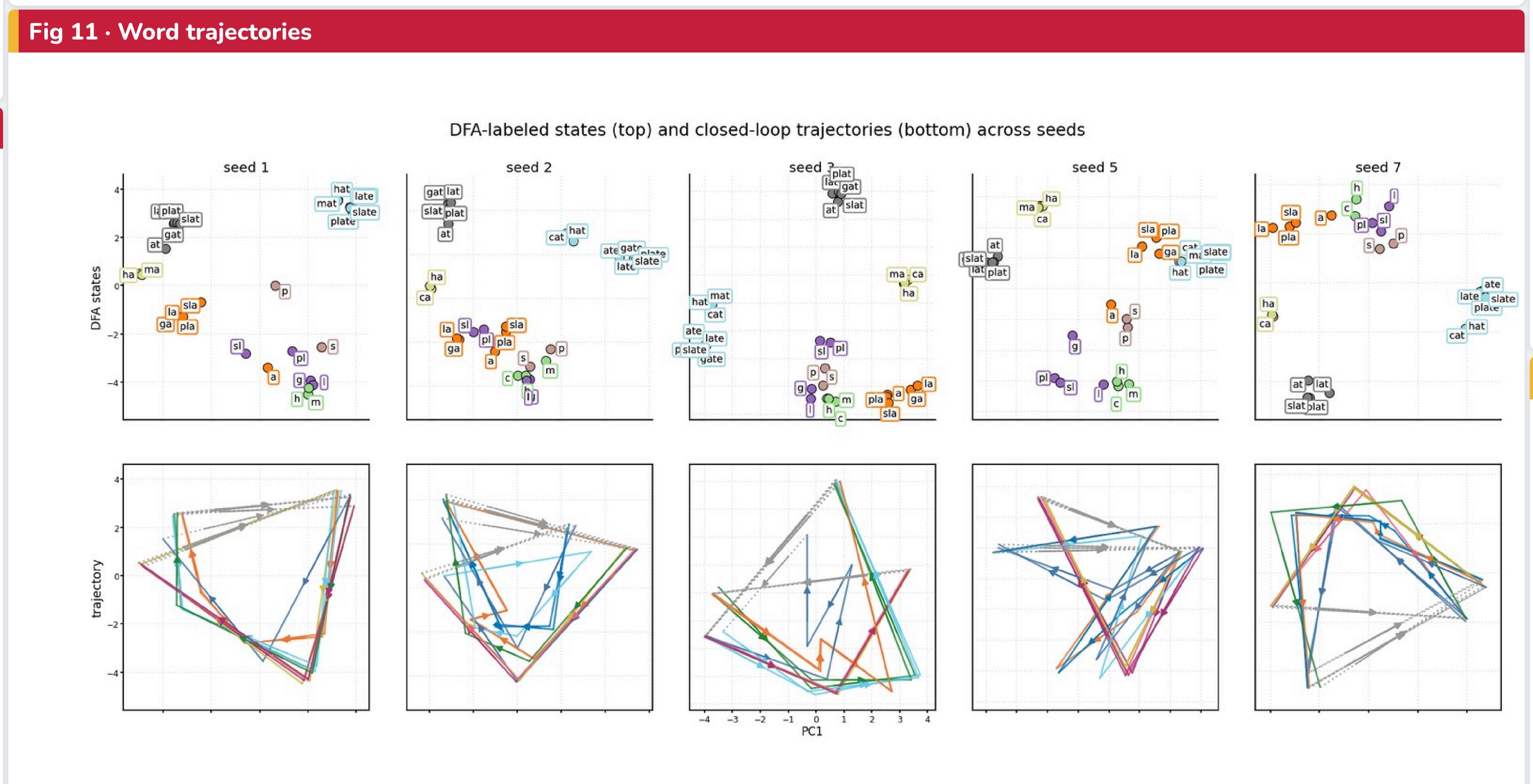
One top unit per feature on a shared corpus window.

Fig 10 · Linear decoding



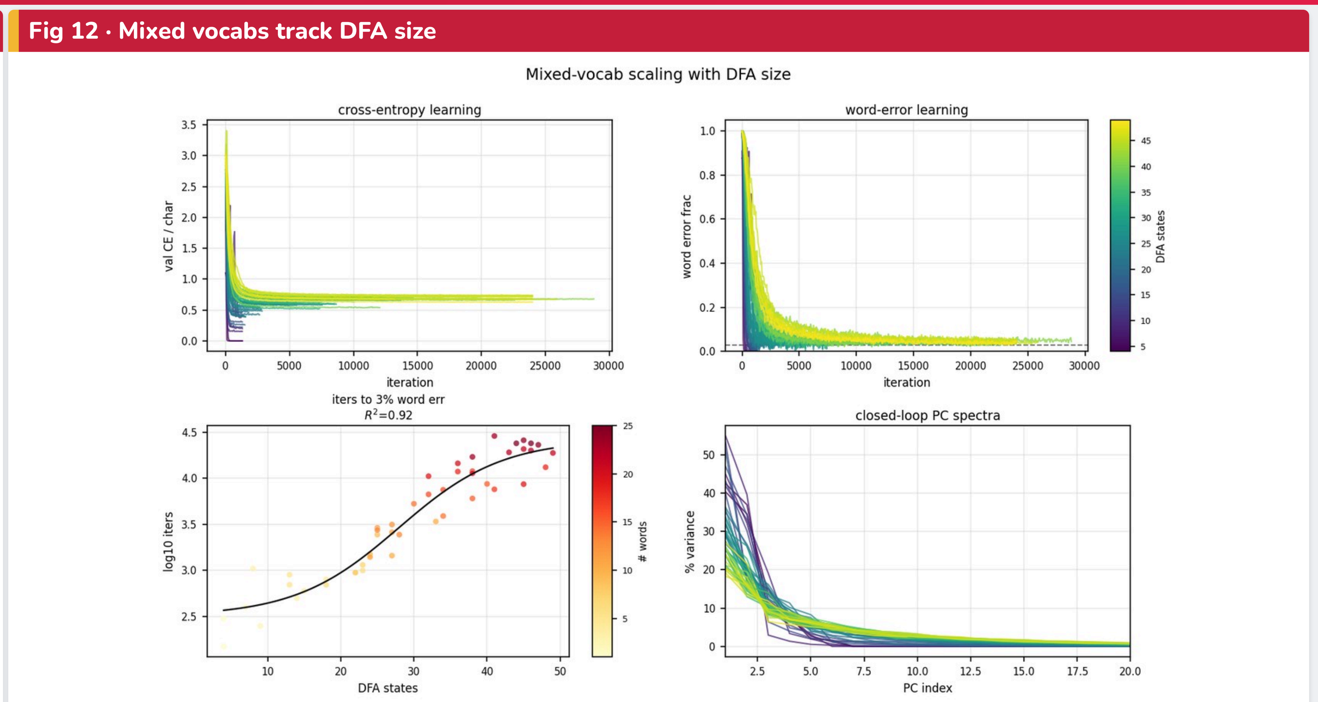
Mean ± std, 6 seeds. DFA/character saturate in a few PCs; word identity needs many more.

Fig 11 · Word trajectories



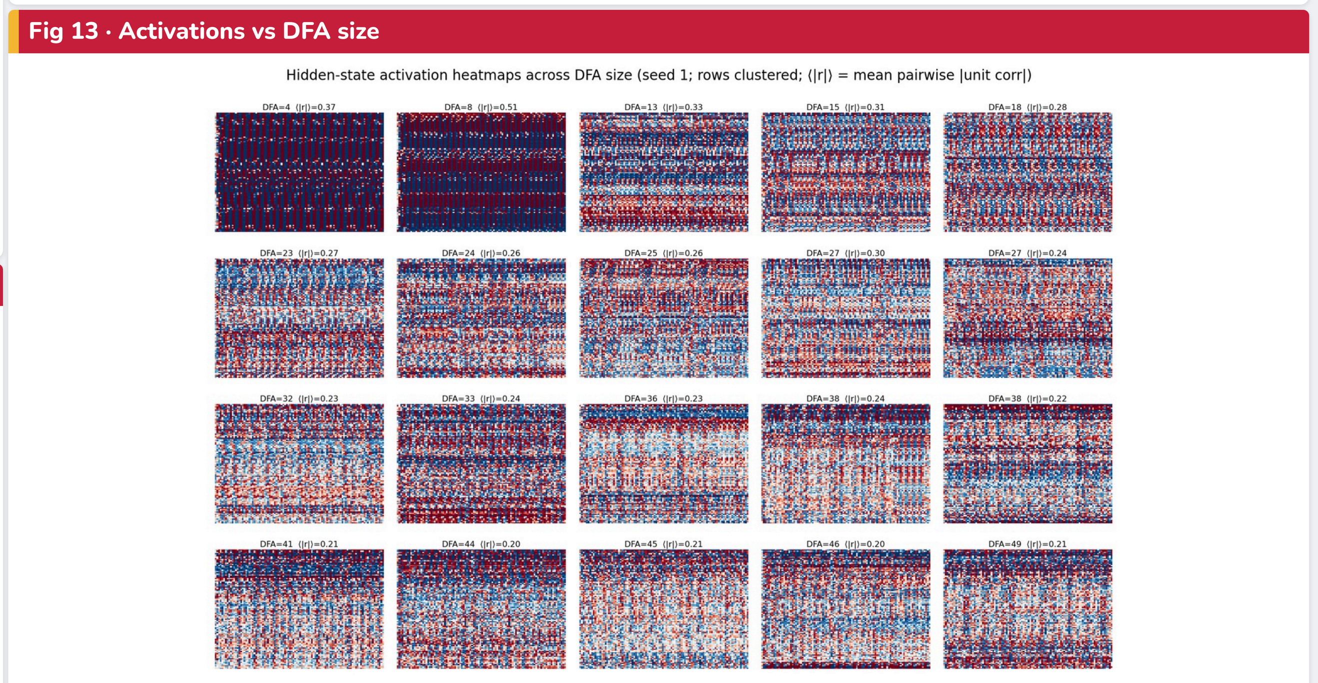
DFA-labeled PCA (top) and closed-loop trajectories (bottom) across five seeds.

Fig 12 · Mixed vocabs track DFA size



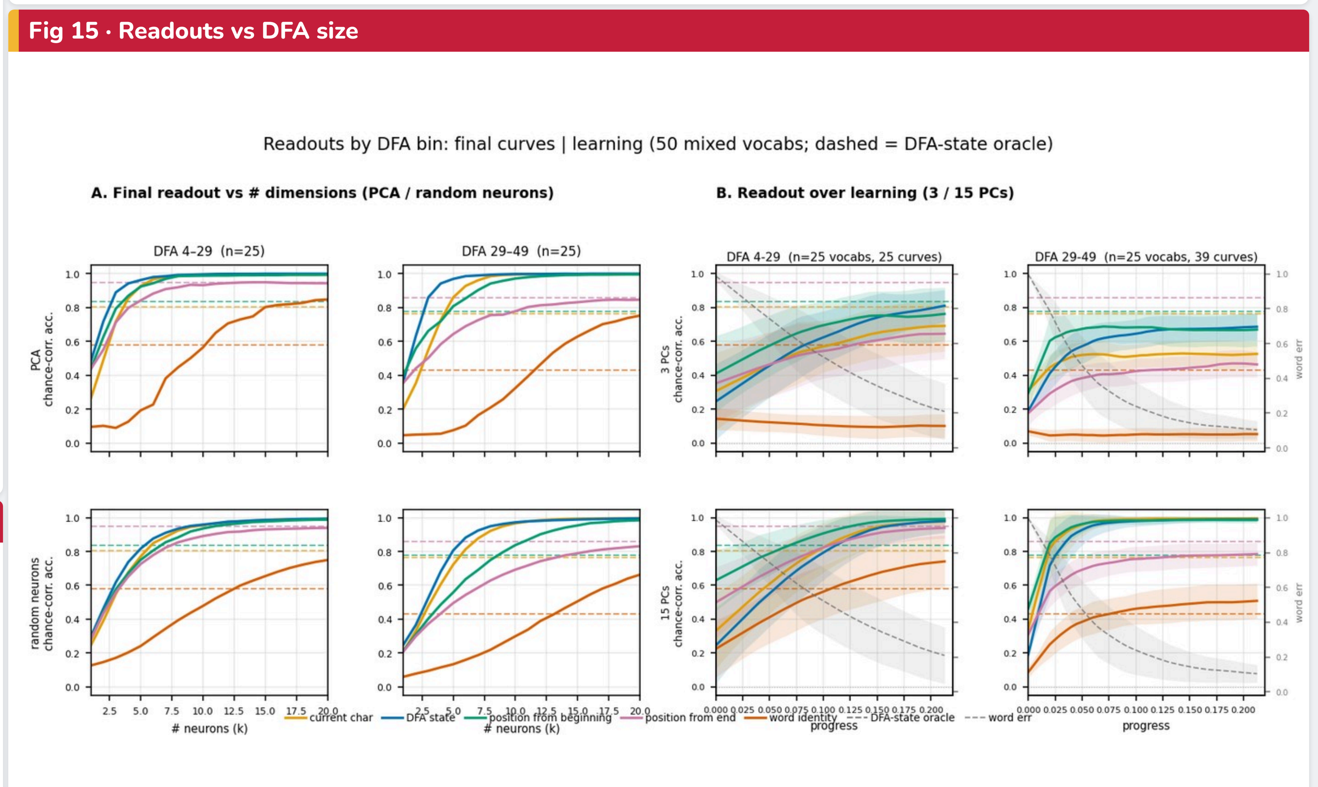
50 mixed vocabs ($H{=}100$). Top: CE and word-error curves by DFA size. Bottom: iters to 3\% WE and closed-loop PC spectra.

Fig 13 · Activations vs DFA size



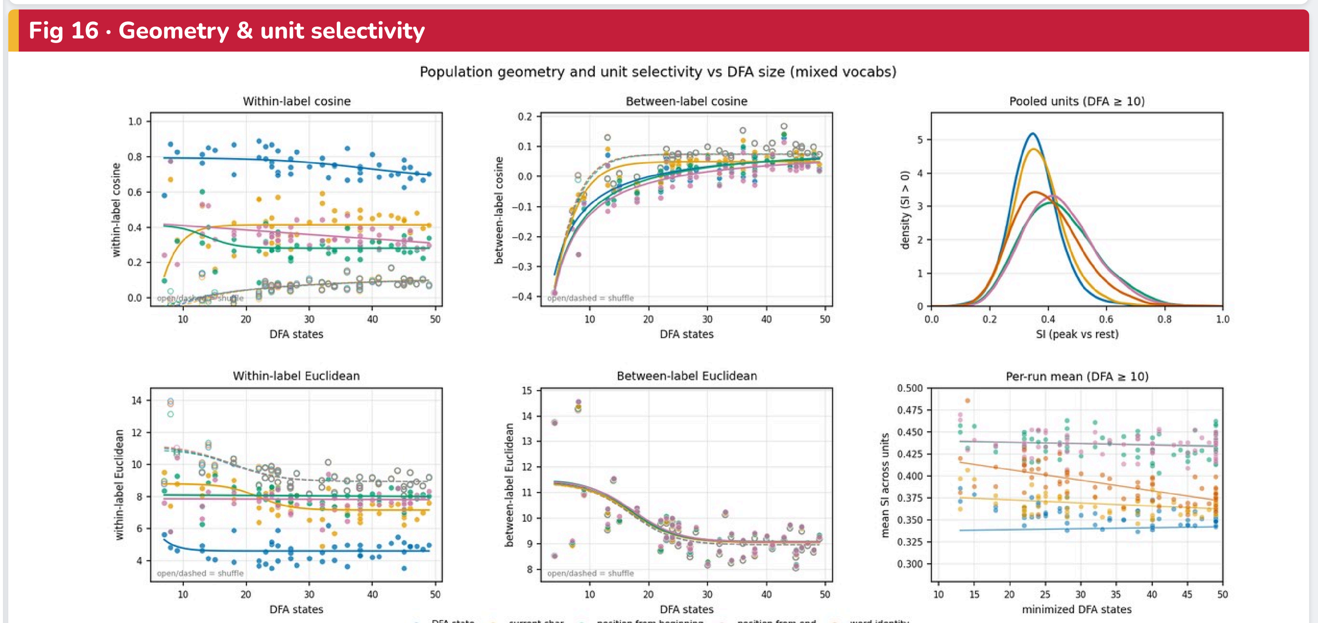
20 mixed vocabs. Low-DFA: coarse bands; larger DFAs: denser, less repetitive rasters.

Fig 15 · Readouts vs DFA size



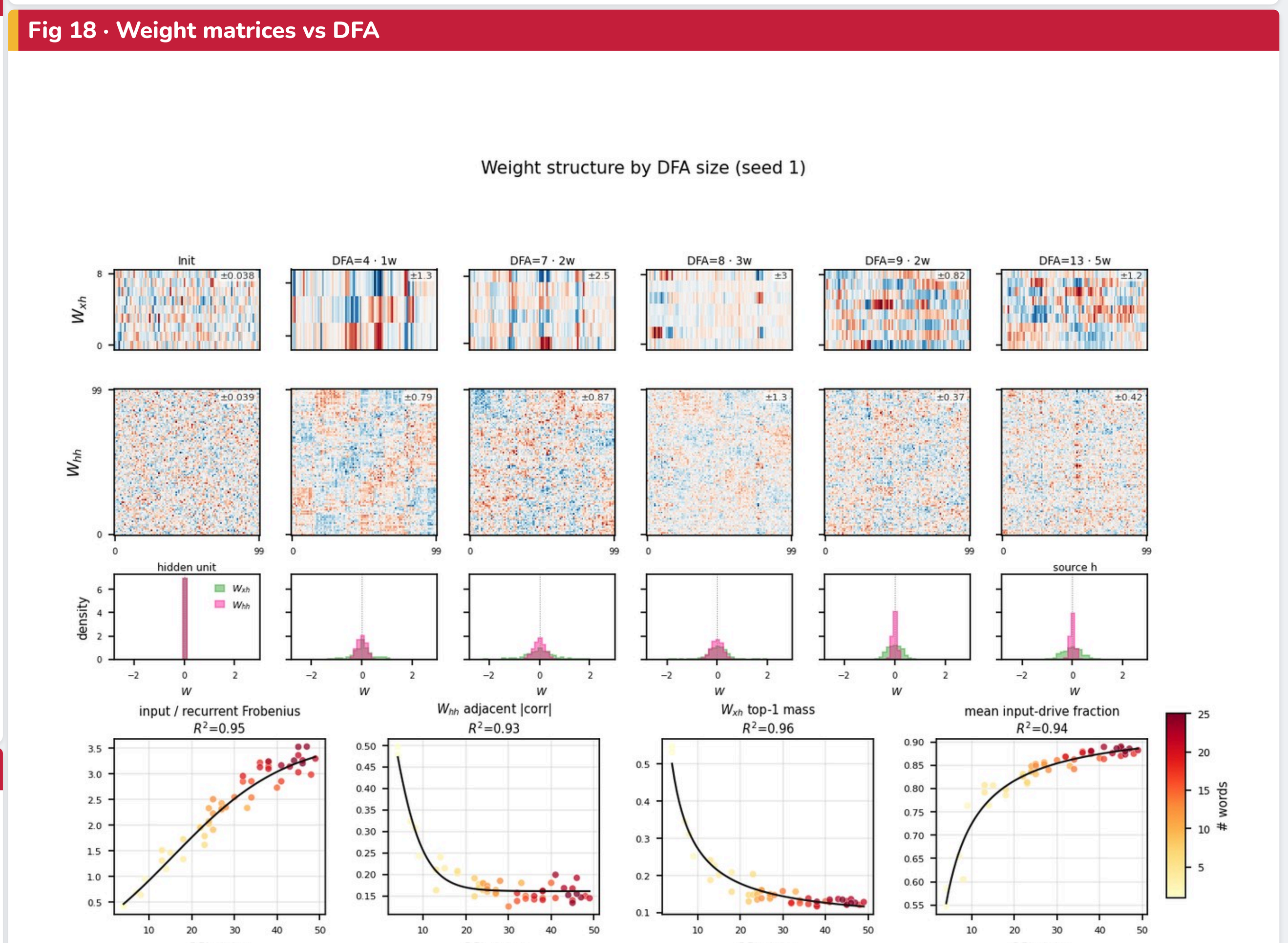
Two DFA bins. Left: final PCA/neuron readout. Right: readout over learning (3/15 PCs). Word identity needs ~15 PCs.

Fig 16 · Geometry & unit selectivity



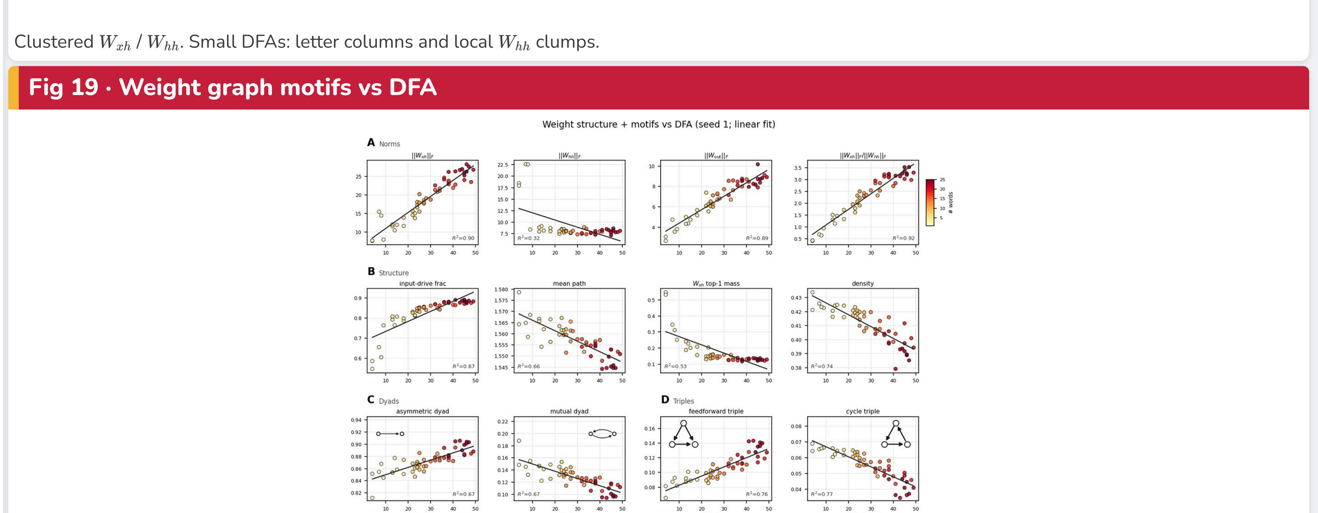
Within/between cosine and Euclidean distance, plus per-unit SI vs DFA size.

Fig 18 · Weight matrices vs DFA



Clustered W_{xh} / W_{hh}. Small DFAs: letter columns and local W_{hh} clumps.

Fig 19 · Weight graph motifs vs DFA



Norms, structure, pairwise and 3-node motifs. Larger DFAs shift feedforward.

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Key references: Saffran, Aslin & Newport (1996); Elman (1990); Aslin, Saffran & Newport (1998); Goldwater et al. (2009); Perruchet & Vinter (1998).